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Spectral Element-Based Seismic Modeling of the Complex Near-Surface with Meter-Scale Heterogeneities.

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Abstract

Seismic exploration in arid regions presents significant challenges due to intense scattering noise generated by highly heterogeneous near-surface layers. These complex geological conditions necessitate the development of advanced processing algorithms capable of accurately characterizing wave propagation phenomena. To systematically investigate near-surface effects, we have developed an efficient and precise forward modeling tool based on the Spectral Element Method (SEM), specifically designed to simulate seismic wave propagation in strongly heterogeneous media. Our implementation incorporates several innovative computational strategies to achieve both accuracy and efficiency. The method capitalizes on SEM's high-order approximation capabilities and its flexibility with unstructured grids, enabling faithful representation of small-scale, high-contrast near-surface inclusions. The computational framework features adaptive mesh refinement, maintaining fine discretization in shallow heterogeneous zones while employing coarser elements in deeper, more uniform strata. Furthermore, we have developed an optimized time-stepping algorithm that automatically determines stable time increments while minimizing computational overhead. This integrated approach results in a robust modeling toolkit that accurately simulates seismic wave propagation through complex near-surface environments characterized by strong heterogeneity and impedance contrasts. The methodology provides an essential foundation for developing and testing advanced processing algorithms aimed at mitigating scattering noise in arid region seismic data.

Introduction

Near-surface geological layers frequently exhibit intricate heterogeneity, which profoundly influences seismic wave propagation and distorts recorded seismograms. A notable example is the Late Jurassic Middle Eastern carbonate strata, characterized by laterally variable lagoon-dominated stromatoporoid-coral facies [1]. Meter-scale geological variations with high impedance contrasts generate geological clutter [2], manifesting as coherent noise known as speckle scattering (Figure 1). Such distortions pose significant challenges in seismic data processing and interpretation. To mitigate these issues, the development of advanced seismic processing algorithms is essential, with an accurate and efficient forward modeling solver serving as a cornerstone.

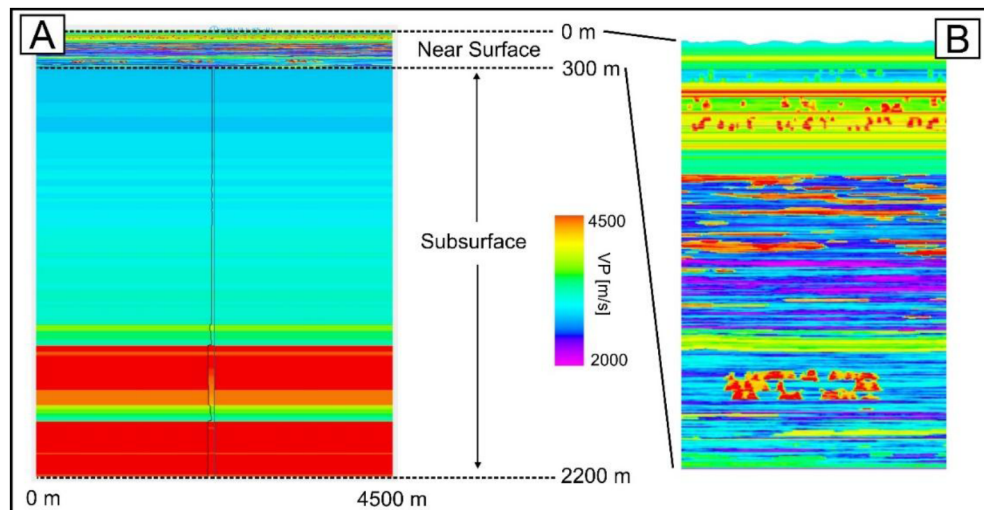


Figure 1—(A) Full P-wave velocity model, including the detailed near-surface and parallel-bedded sub-surface zones. (B) A zoom-in of the near-surface zone exemplifying facies-driven P-wave heterogeneity that forms geologic clutter

This study aims to develop a numerical scheme capable of accurately simulating seismic wave propagation in highly heterogeneous near-surface layers with complex topography. Achieving this objective requires the selection of an appropriate numerical method and the optimization of its computational performance. An ideal method should accommodate unstructured meshes to leverage adaptive refinement in high-contrast velocity zones while maintaining a high convergence order to resolve fine-scale wavefield complexities.

Numerous numerical methods have been developed for seismic wave propagation, each with distinct advantages and limitations depending on the application [3]. The finite difference method (FDM) remains widely used due to its simplicity [4], yet its reliance on structured grids restricts its applicability in near-surface scenarios with irregular topography. While mimetic finite differences can incorporate surface topography [5], they struggle with localized mesh refinement in heterogeneous zones. Additionally, conventional FDM suffers from low-order accuracy, necessitating excessive grid points—over 80 per Rayleigh wavelength—to achieve acceptable error margins in benchmark problems [6].

The finite volume method (FVM) supports unstructured meshes [7], but its extension beyond second-order accuracy is non-trivial. In contrast, the finite element method (FEM) excels in handling complex geometries and heterogeneous media [8], though low-order implementations are prone to numerical dispersion [9]. High-order FEM formulations alleviate this issue [10,11], yet alternative approaches may offer superior efficiency.

Among high-order methods, the discontinuous Galerkin (DG) approach combines FEM's geometric flexibility with enhanced features such as high-order accuracy and strict energy conservation. However, studies [12] suggest that DG imposes stricter stability constraints than the spectral element method (SEM) for both acoustic and elastic wave equations.

SEM emerges as a particularly suitable choice for near-surface seismic modeling due to its ability to operate on unstructured grids, adapt to topographic variations, and support localized refinement. Its spectral convergence properties allow for accurate wavefield resolution with relatively few points per wavelength [13]. Moreover, SEM exhibits more lenient stability conditions than high-order DG, enhancing computational efficiency.

This paper is organized as follows: Section 1 introduces the problem and objectives. Section 2 outlines the geological model and problem formulation. Section 3 details the SEM framework and strategies for optimizing its efficiency. Section 4 presents synthetic seismograms generated for the proposed geological model. Finally, conclusions are drawn based on numerical experiments.

Model

The Late Jurassic carbonate strata of the Middle East exhibit significant meter-scale heterogeneity in stromatoporoid-coral facies [14,15], characterized by complex three-dimensional morphologies and patchy distributions [16]. This geological heterogeneity generates substantial seismic speckle noise in Arabian land seismic datasets [17]. To accurately represent these features, we developed a 2D outcrop-based model incorporating P-wave velocity, S-wave velocity, and density distributions that faithfully capture the near-surface complexity of this facies.

Our model spans 4.5 km laterally and extends to 2.2 km depth, discretized at 0.3 m resolution (Figure 1A). The domain is divided into two distinct zones: (1) a highly detailed near-surface zone (0-300 m) containing realistic meter-scale heterogeneity of stromatoporoid-coral facies and associated background strata, and (2) a simplified subsurface zone (300-2200 m) implemented as a layer-cake model with properties derived from reference [18].

The near-surface zone was modeled using a process-based facies modeling approach [17] that simulates the hierarchical scaling relationships observed in outcrop. This method realistically recreates the morphology of stromatoporoid-coral buildup clusters and their flanking strata, constrained by predefined stratigraphic surfaces. Unlike mathematical clutter generation techniques, this approach produces geologically plausible facies arrangements.

Seismic properties were assigned to facies types based on published measurements [16]. Stromatoporoid-coral buildups were characterized by average properties of $V_p = 4500$ m/s, $V_s = 2600$ m/s, and density = 2.4 g/cm³, while background strata were assigned $V_p = 2400$ m/s, $V_s = 1730$ m/s, and density = 2.2 g/cm³. The model was completed by superimposing a realistic topographic profile on the near-surface zone. The final model (Figure 1B) successfully captures both facies-driven and topography-induced heterogeneity contrasts representative of Late Jurassic Middle Eastern carbonate environments.

Method

In this paper, we analyze seismic response from high heterogeneous near surface inclusions with non-trivial topography by employing the elastic wave equation.

Equation of motion

The propagation of seismic waves can be effectively represented using the elastic wave equation:

$$\rho \ddot{\mathbf{u}} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{F} \quad (1)$$

with the background medium's elastic properties assumed isotropic. The elastic properties of the background medium are presumed to be isotropic. This is characterized by Hooke's law, establishing a linear relationship between the stress and strain tensors:

$$\sigma_{ij} = \lambda \epsilon_{ll} \delta_{ij} + 2\mu \epsilon_{ij} \quad ; \quad \epsilon_{ij} = \frac{1}{2} (\nabla_j u_i + \nabla_i u_j) \quad (2)$$

The presence of a clutter zone necessitates high-order convergence numerical methods to accurately represent velocity anomalies that are high in contrast and small in scale. We propose employing the Spectral Element Method for this task. SEM's spatial discretization is mathematically derived from the finite element method, beginning with its weak formulation. Using the formalism of the Galerkin method, we define the new vector of discretized displacement field $\mathbf{d}$ whose components are the linear combinations of base functions $N_n(\mathbf{x})$:

$$\mathbf{d}(t, \mathbf{x}) = \sum \mathbf{D}_n(t) N_n(\mathbf{x}) \quad (3)$$

By substituting expressions (2) and (3) into equation (1) for the continuous displacement field $\mathbf{u}$, we obtain a residual:

$$R = \rho \ddot{\mathbf{d}} - \nabla \cdot \boldsymbol{\sigma} - \mathbf{F} \quad (4)$$

We force this residual to be orthogonal to the linear space spanned by functions $N_n(\mathbf{x})$. To achieve this, we multiply the expression (7) by each basis function $N_m(\mathbf{x})$, integrate over the computation domain V , and enforce that the resulting integral equals zero:

$$\int_V N_m (\rho \ddot{\mathbf{d}} - \nabla \cdot \boldsymbol{\sigma} - \mathbf{F}) dV = 0 \quad (5)$$

After applying Green's theorem to (6), we can rewrite a weak formulation and get:

$$\int_V (\rho \ddot{\mathbf{d}}) N_m dV - \int_V \boldsymbol{\sigma} (\nabla N_m) dV + \int_S (\boldsymbol{\sigma} \cdot \mathbf{n}) N_m dV + \int_V \mathbf{F} N_m dV = 0 \quad (6)$$

In this study, the relevant free surface boundary condition is expressed as $\boldsymbol{\sigma} \cdot \mathbf{n} = 0$, leading to the elimination of the third integral over surface S in equation (6). The accuracy of the approximated solution hinges on selecting appropriate basis functions $N_m(\mathbf{x})$. Unlike the classical Finite Element approach which uses low-degree polynomials, the Spectral Element Method (SEM) employs high-degree Lagrange polynomials for a higher order approximation. These polynomials, forming the basis functions, interpolate the solution at Gauss-Lobatto-Legendre (GLL) quadrature rule nodes. The GLL nodes ensure optimal numerical integration precision, crucial for computing the integrals in equation (6). This design results in high-order convergence, vital for accurately modeling strongly dispersive ground roll waveforms in near-surface clutter zones.

Spatial discretization

The Spectral Element Method (SEM) retains the geometric flexibility of finite elements but employs quadrilateral (rather than triangular) elements in two-dimensional applications. While quadrangular meshes inherently support complex terrain geometries, their true adaptability is realized through the use of deformed elements (Fig. 2). To accurately represent free-surface topography, we employ curvilinear elements enriched with additional edge-center nodes, ensuring precise geometric fidelity.

Like other finite element-based approaches, SEM facilitates wave propagation modeling on unstructured grids. Given that small-scale near-surface heterogeneities and topographic features demand fine spatial discretization, we implement local mesh refinement in regions requiring high resolution while maintaining coarser elements in deeper, more homogeneous layers (Fig. 2). This strategy ensures that the mesh captures all essential structural details while significantly reducing the total cell count compared to a uniformly refined grid with equivalent geometric accuracy.

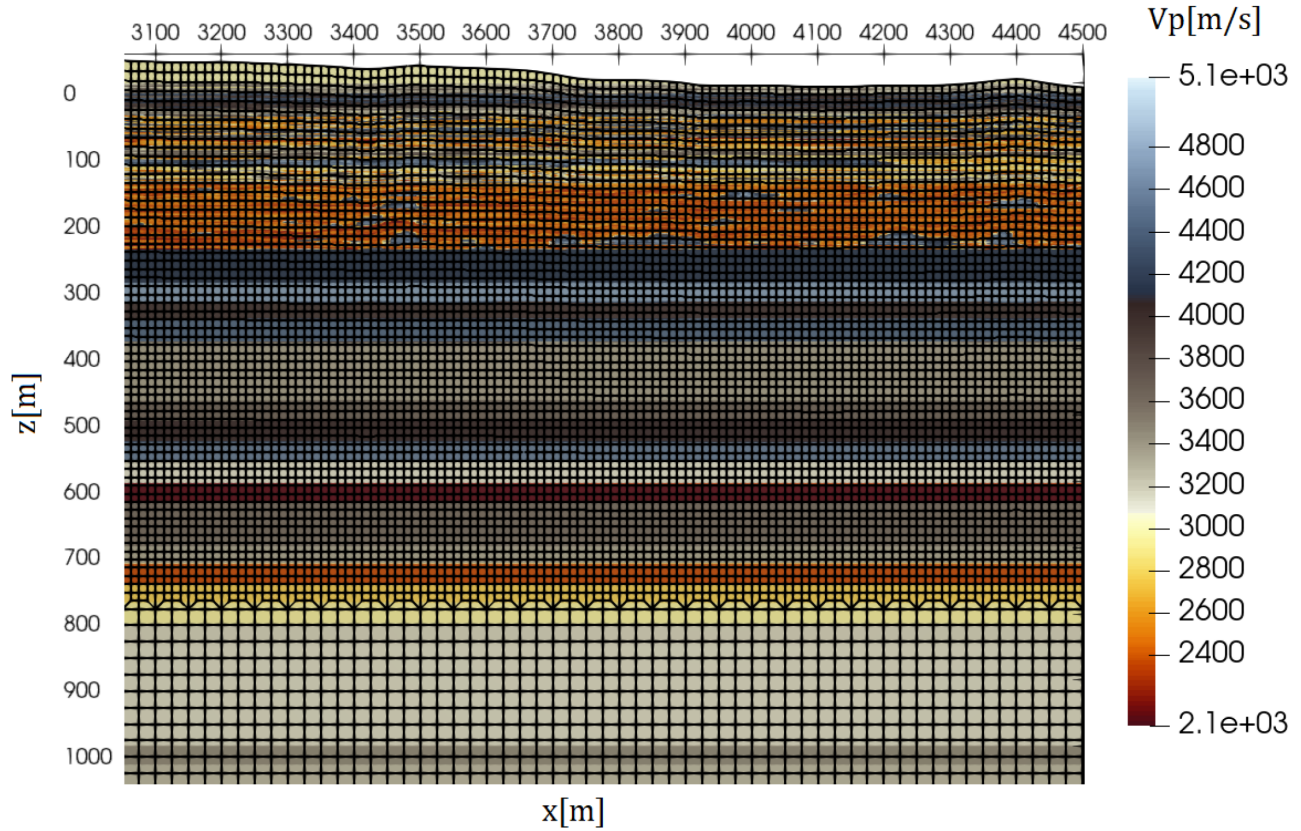


Figure 2—Mesh refinement strategy: localized grid densification in near-surface high-contrast zones (e.g., anomalies and topography) contrasted with sparser discretization in deeper, more uniform regions.

Temporal discretization

The numerical integration, performed using the Gauss-Legendre-Lobatto quadrature, allows equation (6) to be reformulated in a matrix form, what leads to mass matrix and stiffness matrix:

$$\mathbf{M}\ddot{\mathbf{d}} + \mathbf{K}\mathbf{d} + \mathbf{F} = 0, \quad (7)$$

The mass matrix for SEM, constructed with GLL integration rule, results in a completely diagonal form, making the time marching scheme to be explicit [19]. The temporal discretization is done by the Newmark method [20]:

$$\begin{aligned} \dot{\mathbf{u}}_{n+1} &= \dot{\mathbf{u}}_n + (1-\gamma)\delta t \ddot{\mathbf{u}}_n + \gamma\delta t \ddot{\mathbf{u}}_{n+1} \\ \mathbf{u}_{n+1} &= \mathbf{u}_n + \delta t \dot{\mathbf{u}}_n + 0.5\delta t^2((1-2\beta)\ddot{\mathbf{u}}_n + 2\beta\ddot{\mathbf{u}}_{n+1}) \end{aligned} \quad (8)$$

The time step is constrained by stability requirements. While the classical Courant-Friedrichs-Lewy condition provides time step limitations for regular rectangular grids, our implementation must account for irregular mesh elements and strong velocity contrasts in heterogeneous media. We therefore employ a more rigorous approach through spectral analysis of the update matrix $\mathbf{A}$, which governs the transition between time steps:

$$\begin{bmatrix} \mathbf{u}_{n+1} \\ \dot{\mathbf{u}}_{n+1} \end{bmatrix} = \mathbf{A} \cdot \begin{bmatrix} \mathbf{u}_n \\ \dot{\mathbf{u}}_n \end{bmatrix} \quad (9)$$

The update matrix $\mathbf{A}$ takes the form:

$$\mathbf{A} = \begin{bmatrix} \mathbf{I} - \delta t^2 \cdot 0.5 \mathbf{M}^{-1}\mathbf{K} & \delta t \mathbf{I} \\ -\delta t \mathbf{M}^{-1}\mathbf{K} + \delta t^3 \cdot 0.25 (\mathbf{M}^{-1}\mathbf{K})^2 & -0.5 \delta t^2 \mathbf{M}^{-1}\mathbf{K} + \mathbf{I} \end{bmatrix} \quad (10)$$

The stability criterion requires $\rho(\mathbf{A}) \leq 1$, where $\rho(\mathbf{A}) = \max\{|\lambda_1|, \dots, |\lambda_N|\}$ is the spectral radius of $\mathbf{A}$. We determine the optimal time step by solving:

$$\rho(\mathbf{A}(\delta t)) - 1 = 0 \quad (11)$$

using Newton's method. The dominant eigenvalue required for this computation is efficiently estimated through the power iteration method, which offers significant computational advantages by operating directly on the existing matrix-vector multiplication framework without requiring explicit assembly of $\mathbf{A}$. This approach automatically incorporates the geometric complexity of unstructured meshes and material property variations, yielding a time step that optimally balances stability requirements with computational efficiency for the specific problem configuration.

Results

We performed seismic wave propagation simulations for the geological model introduced earlier, characterized by complex topography and strong near-surface heterogeneity (Fig. 1). The model spans 2200 m in depth and 4500 m laterally, with high-contrast inclusions concentrated within the uppermost 250 m. A vertical force source, represented by a 30 Hz Ricker wavelet, was applied at the surface. Given the near-surface shear wave velocity of 1000 m/s, which yields an estimated Rayleigh wave velocity of 700 m/s, the minimum wavelength for 30 Hz signals is approximately 23 m. Following established guidelines [24,25], we employed an 8th-order spectral element scheme with one element per wavelength to ensure accurate wavefield resolution.

The wavefield snapshots at 0.4 s and 0.6 s (Fig. 3), superimposed on the velocity model, reveal significant scattering effects induced by near-surface heterogeneities. Much of the seismic energy undergoes multiple scattering and becomes trapped in the shallow layers, interfering with deeper reflections. This phenomenon is further evident in the synthetic seismograms (Fig. 4a), which exhibit distinct noise patterns.

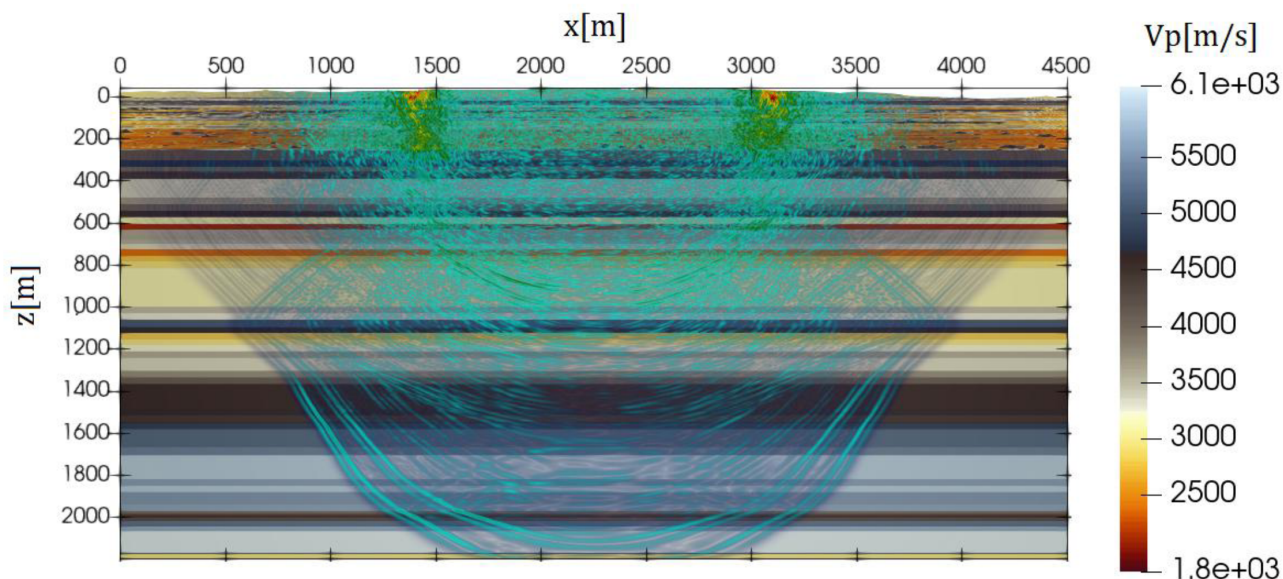


Figure 3—The magnitude distribution at 0.4 sec., shown on a density heat-map

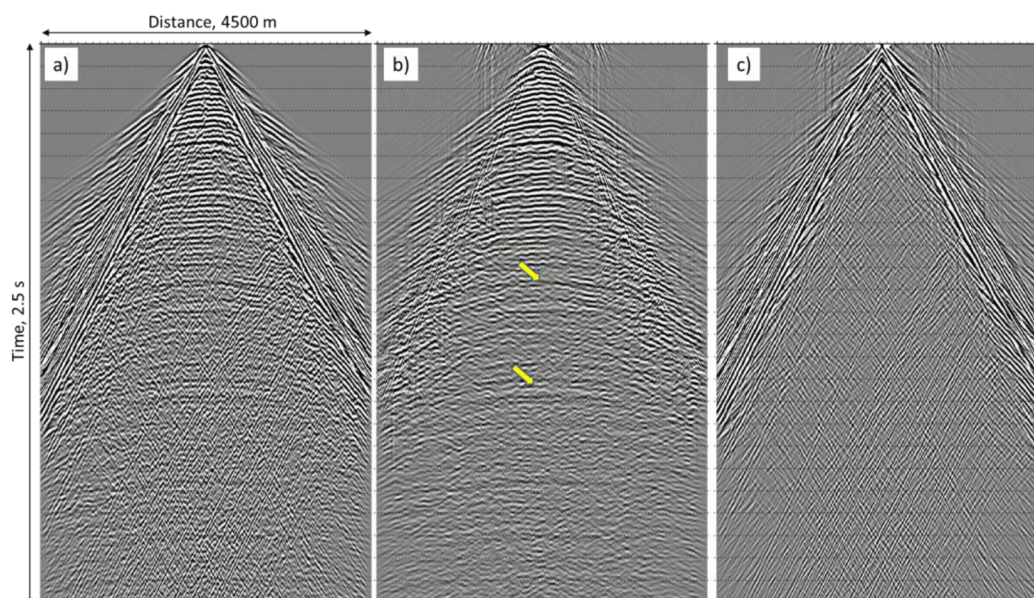


Figure 4—The vertical component (v_z) seismogram reproduces reflections with characteristic speckle noise. The receivers are positioned along the surface of the numerical domain, aligned with the topography. They extend along the X-axis from 5 to 4495 meters, with a spacing of 10 meters between each receiver, resulting in a total of 450 receivers. The raw computed data is shown in a). The result of filtering in F-K domain is shown in b), while the removed linear noise is presented in c). Note the strong phase distortions of deeper reflection events marked by yellow arrows caused by small-scale high contrast heterogeneities in the near-surface.

The recorded data contain significant diffraction energy caused by surface wave interactions with small-scale heterogeneities, appearing as linear dipping noise across the seismic gather. Application of an f-k filter successfully removed this coherent noise component (Fig. 4b), with the extracted noise shown in Fig. 4c. However, deeper reflection events in Fig. 4b still show strong distortions caused by multiplicative phase perturbations, consistent with the seismic speckle noise model [26]. These persistent distortions represent more complex wavefield interactions that cannot be eliminated through conventional filtering approaches.

These results demonstrate both the challenging nature of land seismic data in heterogeneous near-surface environments and the capability of our modeling approach to accurately capture these subtle yet critical wave propagation effects.

Conclusions

We have developed an advanced seismic modeling tool based on the Spectral Element Method, incorporating optimized time discretization and unstructured meshing to accurately simulate wave propagation in complex near-surface environments. Our approach enables precise data interpolation in geologically challenging settings characterized by strong heterogeneity and topographic variations. Numerical simulations using realistic velocity models (Figure 1) successfully reproduced wavefield patterns and noise characteristics comparable to field observations, including the distinctive speckle noise patterns that obscure seismic reflections in actual survey data.

Current research focuses on further enhancing the geological fidelity of our models through the incorporation of viscoelastic attenuation and anisotropic properties. These developments will enable the generation of synthetic datasets that more closely approximate field acquisition conditions. Such improvements are crucial for advancing signal processing techniques, particularly in developing effective methods for speckle noise suppression and reflection event enhancement in seismic data interpretation.

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